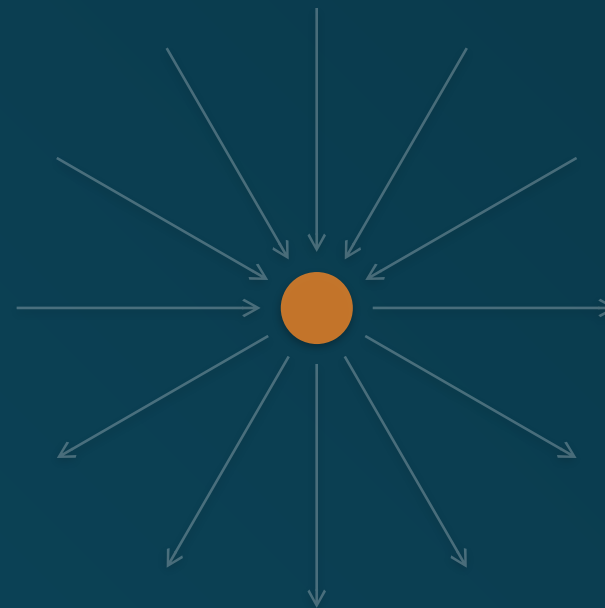


TOPIC 14

Temperature

温度



CIE 9702 · A Level Physics A2 · Paper 4

剑桥国际物理 A2 分章节讲义 · 中英双语

Learning Outcomes

学习目标

LEARNING OUTCOMES

学习目标

By the end of this topic, you will
be able to...

14.1 Thermal equilibrium 热平衡

- Energy is transferred from a region of higher temperature to one of lower temperature 热量自发从高温传向低温
- Regions at equal temperature are in thermal equilibrium 温度相等即热平衡

14.2 Temperature scales 温标

- Physical properties varying with temperature: density of a liquid, volume of a gas at constant pressure, resistance of a metal, e.m.f. of a thermocouple
- Thermodynamic scale does not depend on any particular substance 热力学温标不依赖特定物质
- Convert between kelvin and °C: $T/K = \theta/^{\circ}\text{C} + 273.15$; absolute zero is 0 K 绝对零度

14.3 Specific heat capacity & specific latent heat 比热容与比潜热

- Define and use specific heat capacity 定义并使用比热容
- Define and use specific latent heat; distinguish fusion and vaporisation 区分融化与汽化潜热

Thermal Equilibrium

热平衡与热传递方向

Thermal energy is transferred from a region of **higher temperature** to a region of **lower temperature**.

热量自发地从高温区域传向低温区域。

Two regions at **equal temperature** are in **thermal equilibrium** — no net transfer of energy occurs between them.

温度相等 $\Rightarrow$ 热平衡，无净热传递。

- Temperature measures the **degree of hotness**, not the amount of thermal energy stored 温度是冷热程度，不是热量多少
- A thermometer reaches thermal equilibrium with the object it measures 温度计与被测物达到热平衡后读数



until thermal equilibrium 直至热平衡: **same temperature T**

Measuring Temperature

测温性质与温度计

Physical property 物理性质	How it varies 变化规律	Thermometer 对应温度计
Density / volume of a liquid 液体密度（体积）	Expands on heating 热胀冷缩	Liquid-in-glass thermometer 玻璃液体温度计
Volume of a gas at constant pressure 定压气体体积	$V \propto T$ (kelvin) 与热力学温度成正比	Constant-volume gas thermometer 气体温度计
Resistance of a metal 金属电阻	Increases with temperature 随温度升高增大	Resistance thermometer 电阻温度计
e.m.f. of a thermocouple 热电偶电动势	Depends on junction temperature difference	Thermocouple 热电偶温度计

- Any property that varies **reproducibly** with temperature can define a scale 任何可重复随温变化的性质都可用作测温性质
- Different thermometers may disagree slightly — only the **thermodynamic scale** is substance-independent 只有热力学温标不依赖物质

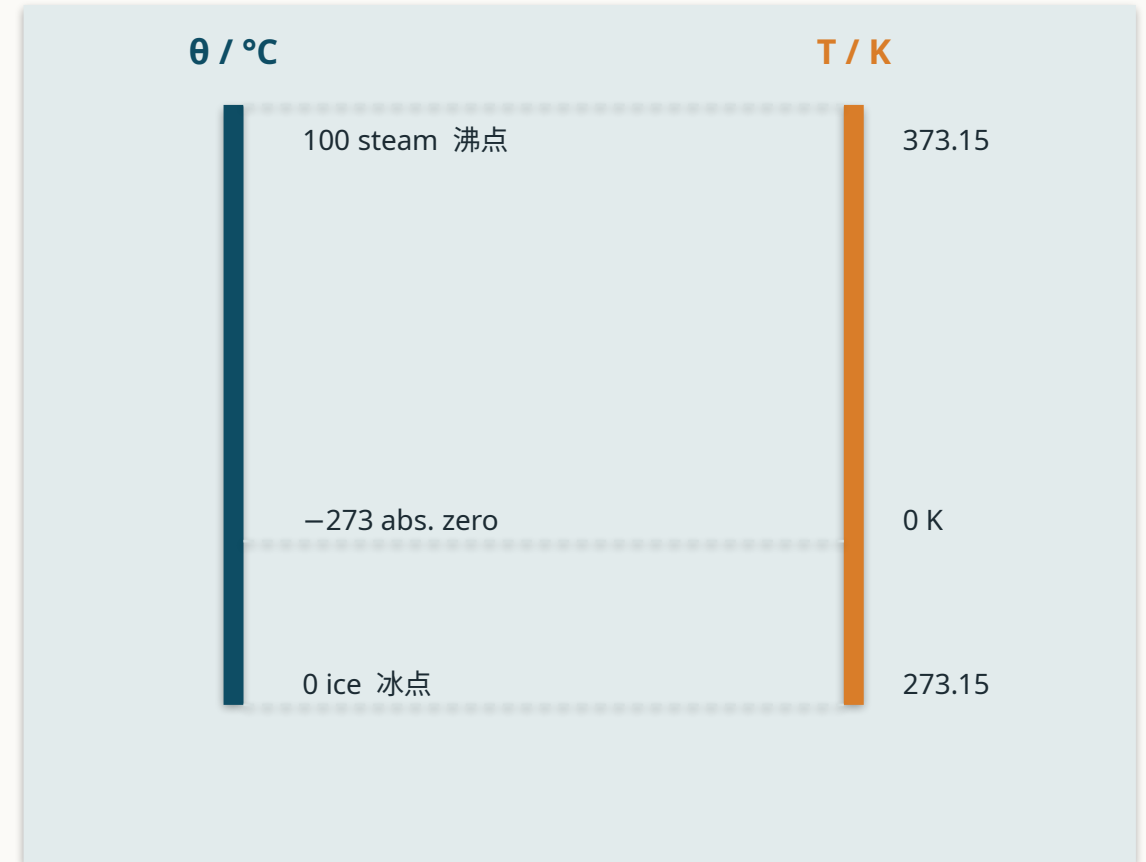
Temperature Scales

摄氏温标与热力学温标

$$T/K = \theta/^{\circ}\text{C} + 273.15$$

kelvin Celsius 换算

- **Absolute zero** 绝对零度 : 0 K = -273.15°C — the lowest possible temperature
- A temperature **difference** of 1 K equals 1°C 温差 $1\text{ K} = 1^{\circ}\text{C}$
- The **thermodynamic (kelvin) scale** does not depend on the property of any particular substance 热力学温标不依赖任何特定物质
- Fixed points 定点 : ice point 273.15 K (0°C), steam point 373.15 K (100°C)



Specific Heat Capacity

比热容

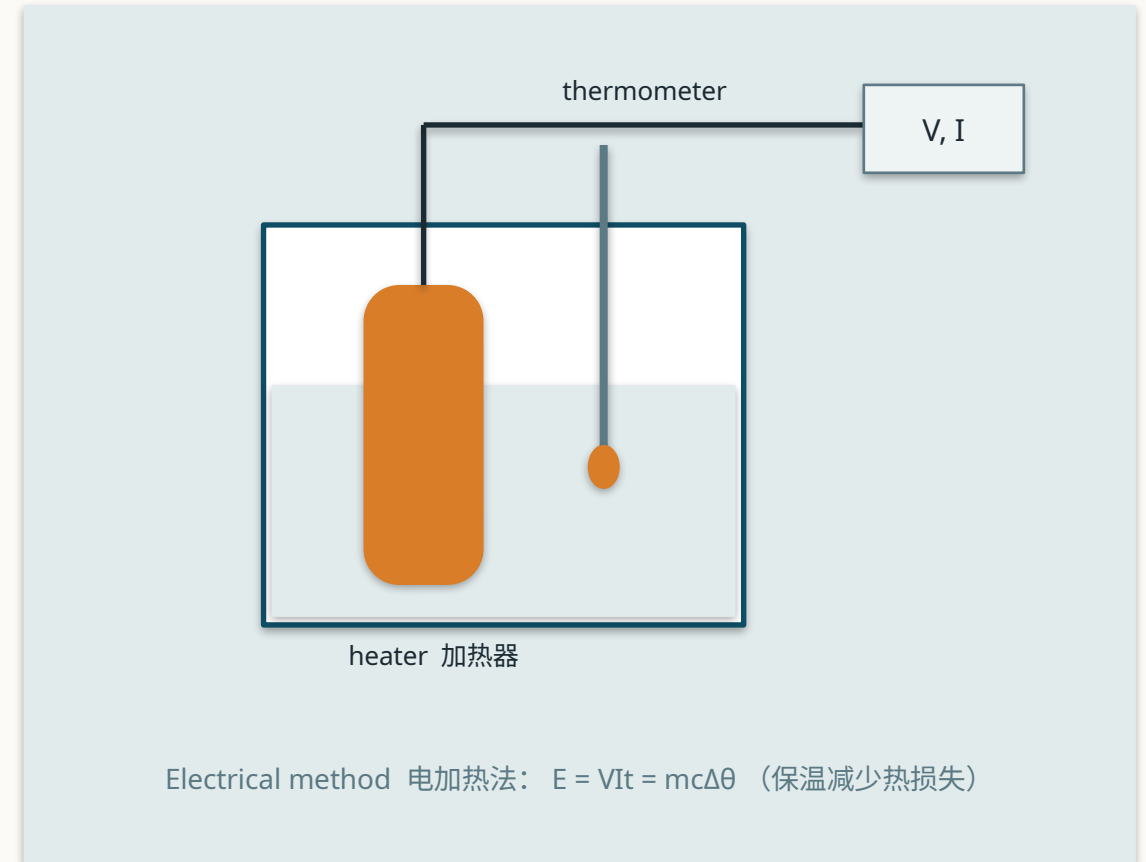
Specific heat capacity c : the energy required per unit mass per unit temperature rise.

单位质量的物质升高单位温度所需吸收的能量。

$$E = m c \Delta \theta$$

c 单位 $\text{J kg}^{-1} \text{K}^{-1}$

- Water has a high ($4200 \text{ J kg}^{-1} \text{K}^{-1}$) — heats and cools slowly 水的比热容大
- Measure with an **electrical heater**: $E = VIt$ supplied to a known mass 电加热法测比热容



Specific Latent Heat

比潜热

Specific latent heat L : the energy per unit mass needed to change state **without any change of temperature.**

单位质量物质在温度不变下完成物态变化所需能量。

$$E = m L$$

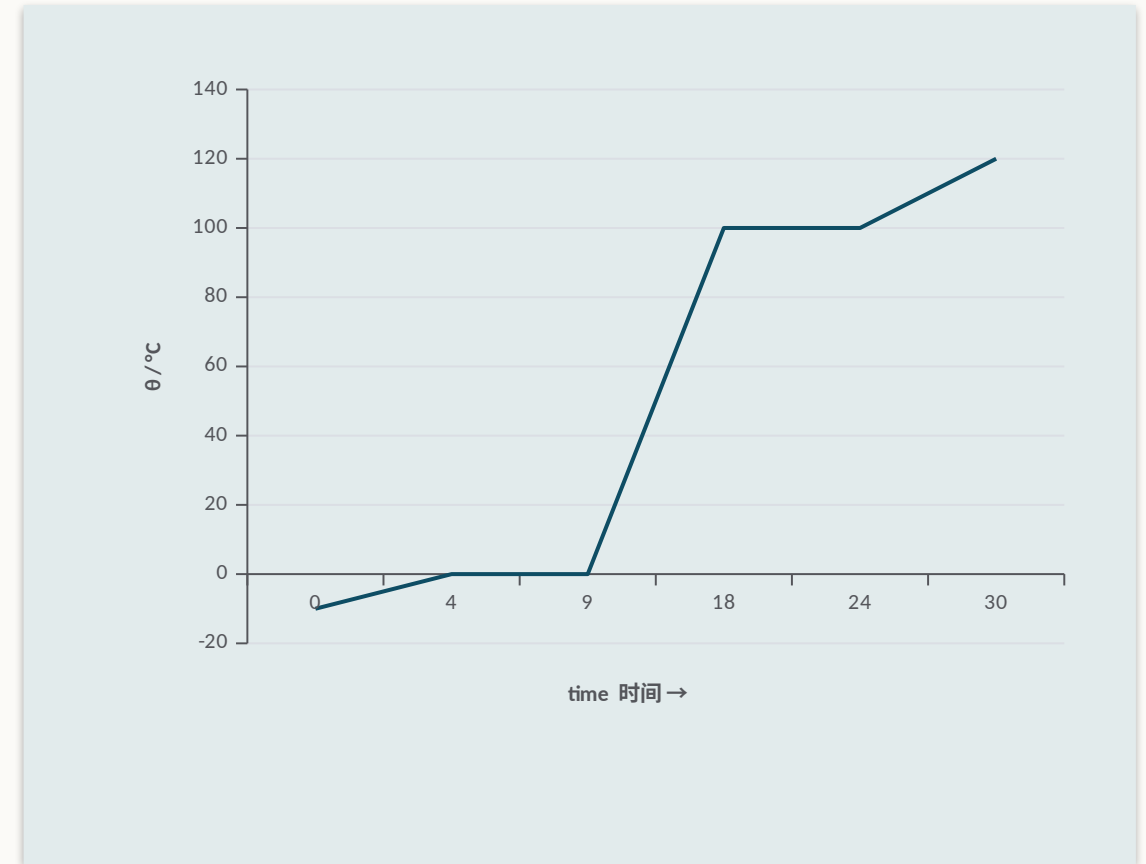
L 单位 J kg^{-1}

- **Fusion** 熔化潜热 L_f : solid $\rightleftharpoons$ liquid (ice/water 330 kJ kg^{-1})
- **Vaporisation** 汽化潜热 L_v : liquid $\rightleftharpoons$ gas (water/steam 2260 kJ kg^{-1})
- During a phase change, energy changes the **potential energy** of molecules, not their kinetic energy — so temperature stays constant 相变时能量改变分子势能而非动能
- $L_v \gg L_f$: separating molecules completely needs far more energy 汽化要彻底克服分子间引力

Heating Curve

从冰到水蒸气的升温曲线

- **Sloping sections** 斜线段: temperature rises, (ice → water → steam 各有斜率)
- **Plateaus** 平台段: phase change at constant temperature, (0 °C 融化、100 °C 汽化)
- Plateau length $\propto$ latent heat needed 平台越长潜热越大
- Heating at constant power $\Rightarrow$ equal time steps = equal energy
恒定功率加热，时间轴即能量轴



Ice heated at constant rate from -10 °C to steam 恒定功率加热

Worked Examples 14A & 14B

例题解析

WORKED EXAMPLE 14A

The temperature of a liquid is measured as $35\text{ }^{\circ}\text{C}$. What is this temperature, to 3 significant figures, on the thermodynamic (kelvin) scale?

- 1.
- 2.

Answer (3 s.f.)

WORKED EXAMPLE 14B

Calculate the thermal energy required to raise the temperature of 590 g of copper from $25\text{ }^{\circ}\text{C}$ to $90\text{ }^{\circ}\text{C}$. ()

- 1.
- 2.

Answer

Worked Example 14C

例题解析

WORKED EXAMPLE 14C

Use Table 14.2 to determine the thermal energy required to melt 78 g of sulfur at its normal melting point.

(Specific latent heat of fusion of sulfur .)

熔点处熔化只需潜热；注意 g $\rightarrow$ kg 换算。

SOLUTION 解答

Step 1 At the melting point, temperature stays constant — use , not .

Step 2 Convert mass: .

Step 3

Answer

Worked Example 14D

例题解析

WORKED EXAMPLE 14D

1 0.30 kg of water at 95 °C is mixed with 0.50 kg of water at 20 °C. Find the final temperature ().

2 15 g of ice at 0 °C is placed in 240 g of drink at 25 °C. Find the final temperature (,).

混合问题先写能量守恒文字方程；冰要先算熔化再算升温。

SOLUTION 解答

Step 1 Word equation 列文字方程 : heat lost by hot = heat gained by cold 热 = 冷 .

Step 2 1 ⇒

Step 3 2 Drink cools; ice melts then warms:

Step 4 Solve: ⇒

Answer 1 2

Class Practice

随堂练习

- Q1** A metal block is heated from $27\text{ }^{\circ}\text{C}$ to $150\text{ }^{\circ}\text{C}$. Determine, to 3 s.f., on the kelvin scale: (a) the temperature of $150\text{ }^{\circ}\text{C}$; (b) the temperature rise. 温标换算。
- Q2** 0.25 MJ of energy raises the temperature of 690 g of water by 86 K . Calculate the specific heat capacity of water. 求水的比热容。
- Q3** Calculate the energy required to (i) melt 76 g of ice at $0\text{ }^{\circ}\text{C}$; (ii) evaporate 76 g of water at $100\text{ }^{\circ}\text{C}$. Hence find how many times more energy evaporation needs than melting. (,)

Answers Q1 (a) 423 K (b) 123 K Q2 Q3 (i) 25.1 kJ (ii) 172 kJ , 约 6.8 倍

Summary

本章小结

KEY FORMULAS 核心公式

$$T / K = \theta / ^\circ C + 273.15$$

kelvin Celsius

$$E = m c \Delta \theta$$

specific heat 比热容

$$E = m L$$

latent heat 潜热

$$E_{lost} = E_{gained}$$

energy exchange 热平衡方程

KEY IDEAS 核心思想

- Heat flows from higher to lower temperature until thermal equilibrium 热量由高温到低温直至热平衡。
- The kelvin scale is absolute and substance-independent; 0 K is absolute zero 热力学温标绝对且与物质无关。
- During a phase change temperature is constant: energy changes molecular PE, not KE 相变时温度不变。

Exam Tips

备考提示

- Temperature **differences** are the same in K and °C, but absolute temperatures in gas/thermo laws **must be in kelvin**. 温差两者相同，绝对温度必须用 K。
- For mixtures, always start with the **word equation**: energy lost by hot object = energy gained by cold object. 先写文字守恒方程。
- With ice in a warm drink, do not forget **two stages**: melting (mL) then warming ($mc\Delta\theta$). 冰的问题别漏熔化段。
- Watch unit traps: g → kg, kJ → J, kW → W. 单位换算是高频失分点。